

Visitor Management

Modern visitor lifecycle for offices, factories, and campuses — from pre-visit invitation, through QR-based gate check-in and identity verification, all the way to hospitality, conference rooms, contact tracing, and audit reports. Built on Frappe & ERPNext.

ERPNext 15

Frappe 15

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Main features

Visitor Lifecycle (invitation → approval → gate → checkout): One Visitor Pass record drives the entire journey. Walk-in passes from reception or portal pre-registration via Visitor Invitation. Visitor type drives the approval workflow lane automatically.

Gate Security & Identity Verification: QR scan at the gate auto-loads the pass. A live photo is captured at the gate, **Matches ID Proof** and **Matches Pass Photo** must be ticked, the system screens against the active **Visitor Blacklist** (by ID number or name+type), issues a badge once items are verified, and records the gate event as an immutable **Security Log** row that can't be edited after save.

Hospitality Management: When a Visitor Pass is approved with any hospitality flag (cab, hotel, factory tour, buggy, greeting, meal), the system auto-creates a **Hospitality Request**. Meal slots are computed against the visit window; cab pick-up/drop times auto-populate from the pass.

Conference Room Booking: If a room is selected on the pass, a **Conference Room Booking** is auto-created. Meeting time is clamped to the room's operating hours and seating capacity is validated.

VIP Protocol: Dedicated **Approved VIP Queue** quick-action at the gate, email notification on pass approval, **protocol notes** carried from the pass through to the gate, pre-allocated meeting room, two-step approval.

Contact Tracing & Audit Trail: Every gate event opens or closes a **Contact Trace Record** for the visitor (visited area, time in, time out, exposure risk). Every lifecycle event is logged in the immutable **Visitor Event Log**.

How to Install

```
cd ~/frappe-bench
bench get-app https://github.com/<owner>/visitormanagement
bench --site <your-site> install-app visitormanagement
bench --site <your-site> migrate
bench build
bench restart
```

Upgrade later:

```
cd ~/frappe-bench/apps/visitormanagement
git pull
cd ~/frappe-bench
bench --site <your-site> migrate
```

Migrations are idempotent — `bench migrate` is safe to run any number of times.

Setup and Use

All system-wide configuration lives in a single doctype: **VMS Settings**. Open the desk Awesome Bar and type "VMS Settings".

Configure VMS Settings

The screenshot shows the 'VMS Settings' interface. On the left is a sidebar with options: Assigned To, Attachments, Share, and a FOLLOW button. The main area is divided into three sections: General, Gate & Security, and Badge Configuration. The General section includes fields for Admin Email (ops.admin@gmail.com), Food Dept Email (catering@gamil.com), Default Checkout Time (18:00:00), and Max Visit Duration (hrs) (12). The Gate & Security section has checkboxes for Require Visitor Photo, QR Scan Required at Gate, Require Item Declaration, and Block Check-In Without Verification. It also features a Blacklist Action dropdown (Block Entry) and a Gate List (Main Gate, Back Gate, VIP Entrance, Loading Bay). The Badge Configuration section includes an Enable Badge checkbox, a description of badge issuance, and a Badge Required For list (Contractor, Candidate, Customer, Supplier, VIP). It also has fields for Badge Prefix (Contractor), Badge Colour (Orange), and Badge Prefix (Candidate).

VMS Settings — General, Gate & Security, and Badge Configuration sections

Field	What to put
Enable Badge	Tick to enable badge generation per visitor
Badge Required For	Comma-separated visitor types that get badges (e.g. Contractor , Customer , Supplier)
Badge Prefix — Contractor / Candidate / Customer / Supplier / VIP	Defaults: CON / CAN / CUS / SUP / VIP
Require Item Declaration	Tick to block pass submission until at least one item row is added
Auto-Cancel Pending Passes (Hours)	After how many hours a Draft pass auto-cancels (0 to disable)
Default Gate	Gate name to fall back to when no auto-assignment rule matches

Click **Save**. The app is now configured.

Configure Workflows

Three approval workflows (Visitor Pass, Hospitality Request, Conference Room Booking) are pre-loaded as fixtures and activated automatically on bench migrate — no setup needed.

Pre-Visit Invitation — invite a visitor in advance

The screenshot shows the 'Visitor Invitation' form for VINV-2026-00006. The form includes fields for Visitor Type (Customer), Visitor Email (demo-invitee-05@example.com), Host Employee (HR-EMP-00007), Visit Date (12-05-2026), Expected Check-in (09:00:00), Expected Check-out (15:00:00), Purpose Of Visit (DEMO Pre-registered visit #6 (Customer)), Created By User (Administrator), Invitation Status (Draft), and Invitation Expires On (12-05-2026 23:59:59). The form also has a sidebar with options: Assigned To, Attachments, Tags, Share, and a FOLLOW button. The top bar shows the breadcrumb 'Visitor Management > Visitor Invitation > VINV-2026-00006' and a search bar.

Visitor Invitation — pre-registration link generated for the visitor

1. Open the desk Awesome Bar and type "Visitor Invitation".
2. Click + **Add Visitor Invitation**.
3. Fill: **Visitor Name, Email, Visit Date, Purpose, Person to Visit**.
4. Click **Save**. The system emails the visitor a token-based portal link.

- The visitor clicks the link, fills the pre-registration form (mobile, ID type, ID number, photo, ID proof scan).
- A **Visitor Pass** is auto-created in Draft state and routed to its approval lane based on visitor type.
- On **Approve** → pass becomes Approved + Submitted, QR code is generated, notification is sent, hospitality request + room booking auto-create if flags are set.

End-to-end takes ~1 minute for the host, ~3 minutes for the visitor.

Walk-in Visitor — visitor is at reception

Visitor Pass — Approved state with QR code and host details

- Open `/app/visitor-pass` and click **+ Add Visitor Pass**.
- Fill: **Visitor Name, Mobile, ID Proof Type + Number, Visitor Type, Person to Visit, Visit Date**.
- Tick hospitality flags if applicable (Meal Required, Cab Required, Hotel Required, etc.).
- Click **Save**. Pass goes to Draft, then **Submit** to start the approval workflow.
- On **Approve** → QR code generated, notification sent, hospitality + room booking auto-created.

End-to-end takes ~5 minutes including approval.

Gate Check-In — visitor arrives

Security Log Check-In — gate photo, identity match, item verification

- Visitor presents QR code at the gate.
- Open `/app/security-log` and click **Scan QR Code**.
- The pass auto-fills — visitor photo, ID details, purpose, host, hospitality flags, items declared.
- Click **Capture Gate Photo** → live photo is uploaded and attached.
- Tick **Matches ID Proof** and **Matches Pass Photo** (both are mandatory).
- Tick each declared item in the **Items Verification** grid as items are checked.
- Click **Save**. Visitor Pass status flips to **Checked-In**, badge is issued, a notification email is sent, a Contact Trace Record opens.

The Security Log is **locked after save** — gate events are immutable audit records.

The system **re-checks the blacklist at the gate** even after pass approval — blacklisting may have happened in between. A blacklisted visitor is refused entry with a security alert email.

Gate Check-Out — visitor leaves

1. Visitor presents QR code at exit.
2. Create a new Security Log with **Event Type = Check-Out**.
3. Live photo + match confirmations (ID + pass photo) are mandatory — same standard as check-in.
4. Click **Save**. Visitor Pass flips to **Checked-Out**, the open Contact Trace Record closes, the badge is marked returned.

Visitors who get blacklisted while still on premises are **allowed to check out** (the goal is to keep them out, not trap them in).

Hospitality flow

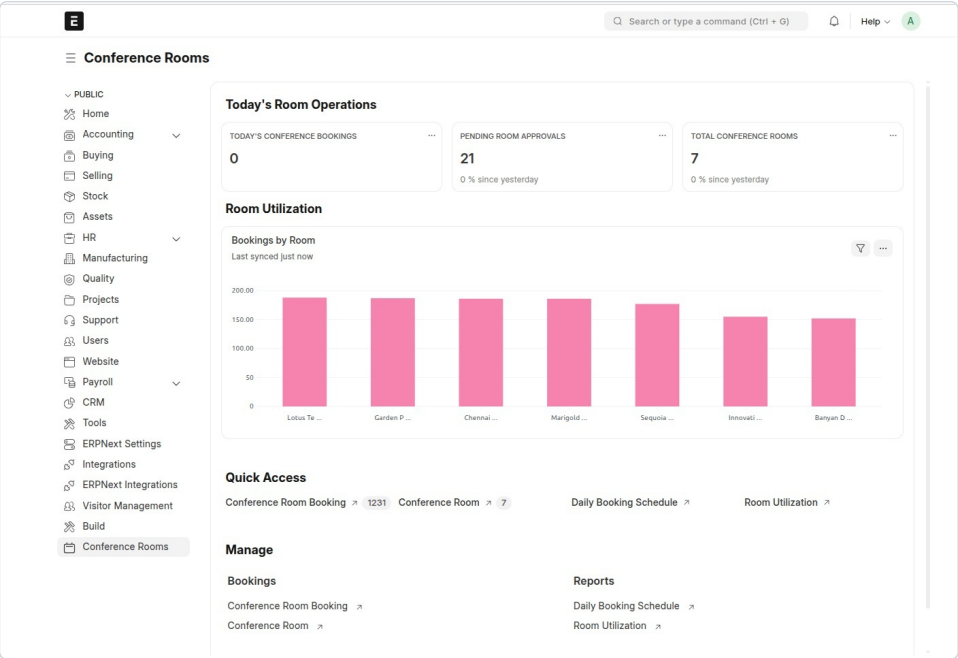
The screenshot shows a web application interface for a 'Hospitality Request'. The breadcrumb trail at the top is 'Visitor Management > Hospitality Request > HOSP-PRA-110526'. The request ID 'HOSP-PRA-110526' is highlighted in green with the status 'Approved'. On the left, there are icons for 'Assigned To', 'Attachments', 'Tags', and 'Share', each with a plus sign. Below these are social media-like icons for 'Like', 'Comment', and 'Share', followed by a 'FOLLOW' button. A status message says 'You last edited this - yesterday' and 'You created this - 3 days ago'. The main form area contains several sections: 'Visitor Pass' with value 'VP-2026-02687'; 'Visitor Name' with value 'Praveen'; 'Visitor Type' with value 'Supplier'; 'Mobile' with value '+91-9874562845'; 'Visit Info' with 'Visit Start Time' '11-05-2026 13:17:33' and 'Visit End Time' '11-05-2026 13:47:42', both with location 'Asia/Kolkata'; 'Dietary & Accessibility' with 'Special Diet' 'None' and 'Accessibility Requirements' 'None' (with a note 'Affects hotel room assignment, buggy selection, tour path'); and 'Meal & Venue' with 'Meal Required' checked, 'Meal Type' 'Lunch', and an empty 'Venue' field. At the bottom is a 'Comments' section.

Hospitality Request — auto-created on pass approval when a hospitality flag is set

When a Visitor Pass is approved with any hospitality flag set, the system creates a Hospitality Request in **Pending Approval**.

1. Open `/app/hospitality-request`.
2. Review meal type, cab pickup/drop, hotel dates, factory tour, greeting type, etc.
3. Assign staff for execution (food dept, cab vendor, hotel partner).
4. Click **Approve** → request becomes Approved + Submitted, an email goes to the assignees.

Conference Room flow



Conference Rooms workspace — today's bookings, pending approvals, room utilization

When a Visitor Pass is approved with a Conference Room selected, the system creates a Conference Room Booking in Pending Approval .

- 1. Open the **Conference Rooms** workspace — shows today's bookings, pending approvals, total rooms, and a **Bookings by Room** chart.
- 2. Open the pending booking and review the time window (clamped to room operating hours) and seating capacity.
- 3. Click **Approve** → booking becomes Approved + Submitted.

Track every visit

Every visitor's journey is captured across three audit doctypes:

Doctype	Captures
Visitor Event Log	Every workflow + lifecycle event — pass created, approved, gate scanned, checked in, checked out, etc. Immutable.
Security Log	Every gate event — Check-In, Check-Out, Gate Transfer, Alert, Badge Collected. Photo at gate, identity match, item verification, gate name. Locked after save.
Contact Trace Record	Per visitor area-by-area movement log — visited area, time in, time out, exposure risk, close contacts. Auto-closes on Check-Out.

Click **Visitor Pass** → **Connections** on any pass to drill into all linked records.

Identity verification at the gate

The Security Log enforces three identity checks for Check-In and Check-Out:

Check	Required when
Live photo at gate	Check-In and Check-Out
Visitor matches the ID proof presented	Check-In and Check-Out
Visitor matches the photo on the approved pass	Check-In and Check-Out

Skipping any blocks the save. The pass photo, ID proof scan, and live gate photo are shown side-by-side in the **Identity Comparison** section.

Multi-Gate Routing

When a visitor arrives without a pre-assigned gate, the Security Log auto-assigns one based on visitor type:

Visitor Type	Default Gate
VIP	VIP Entrance
Supplier	Loading Dock
Contractor	Back Gate
Candidate	Main Gate
Customer	Main Gate

This can be overridden manually on each Security Log or globally via **VMS Settings → Default Gate**.

Visitor Blacklist

Field	What to put
Visitor Name	Person's name (used as fallback when ID number is unknown)
ID Proof Type	Aadhaar / PAN Card / Passport / Driving License
ID Proof Number	Primary lookup key (auto-named series VB-YYYY-#####)
Reason	Mandatory — why this person is blocked (audit/compliance requirement)
Is Active	Tick to enforce the block. Uncheck to suspend the block without deleting the record.

The blacklist is **enforced at three points**: when a Visitor Pass is submitted, when a Security Log Check-In is saved, and when a QR code is scanned at the gate.

A match by ID Proof Number blocks first; if no number is on the blacklist row, the system falls back to a case-insensitive match on Visitor Name + ID Proof Type.

Reports

Seven query reports are pre-installed:

Report	Source DocType	Use
Active Visitors	Visitor Pass	Visitors currently on premises
Daily Visitor Log	Visitor Pass	Day-by-day pass log with status
Gate Wise Count	Security Log	Check-in / check-out / inside / pending counts per gate
Visitor Identity Match Report	Visitor Pass	Audit of identity verification at the gate
Daily Hospitality Schedule	Hospitality Request	Today's cab / hotel / tour / greeting schedule
Daily Booking Schedule	Conference Room Booking	Today's room bookings
Room Utilization	Conference Room Booking	Per-room utilization summary

Limitations

- **Refunds / cancellations** of issued badges are not automated. A pass can be Cancelled via the workflow; badges are revoked on Check-Out only.
- **Recurring visitors** are tracked one Visitor Pass per visit. There is no cross-visit aggregation (e.g. "contractor X visited 12 times this year" requires a custom report on top of the Visitor Pass table).
- **NDA / data-retention purge** is not implemented. Visitor records remain until manually deleted.
- **Items-out reconciliation** (verifying the visitor leaves with the same items they brought in) is not enforced — only items-in is

verified.

- **Currency / localisation** — the app is locale-neutral, but date formatting follows the site's Frappe locale.

Dependencies

- Frappe v15
- ERPNext v15
- HRMS v15 (Employee doctype used for visitor host links)
- Python 3.10+
- MariaDB 10.6+ with InnoDB

License

MIT — see `license.txt`.

Built with  by  **Finstein**